COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S4	**SECTION:**	CPE21S4
DATE PERFORMED:	15/09/2026	**DATE SUBMITTED:**	15/09/2026
NAME:	Jullian Edrain R. Sevilla	**INSTRUCTOR:**	Engr. JIMLORD QUEJADO

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?
ILO B
Problem 1: Can we sort the left sub list and right sub list from the partition method in quick sort using other sorting algorithms? Demonstrate an example.

- Yes, we can because after splitting, the left and right sides are totally separate from each other, so any sorting method works on both sides.

Problem 2: Suppose we have an array which consists of {4, 34, 29, 48, 53, 87, 12, 30, 44, 25, 93, 67, 43, 19, 74}. What sorting algorithm will give you the fastest time performance? Why can merge sort and quick sort have O(N • log N) for their time complexity?

- Quick Sort is the fastest. Both algorithms keep cutting the list in half over and over, so they share the same O(N·log N) speed.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

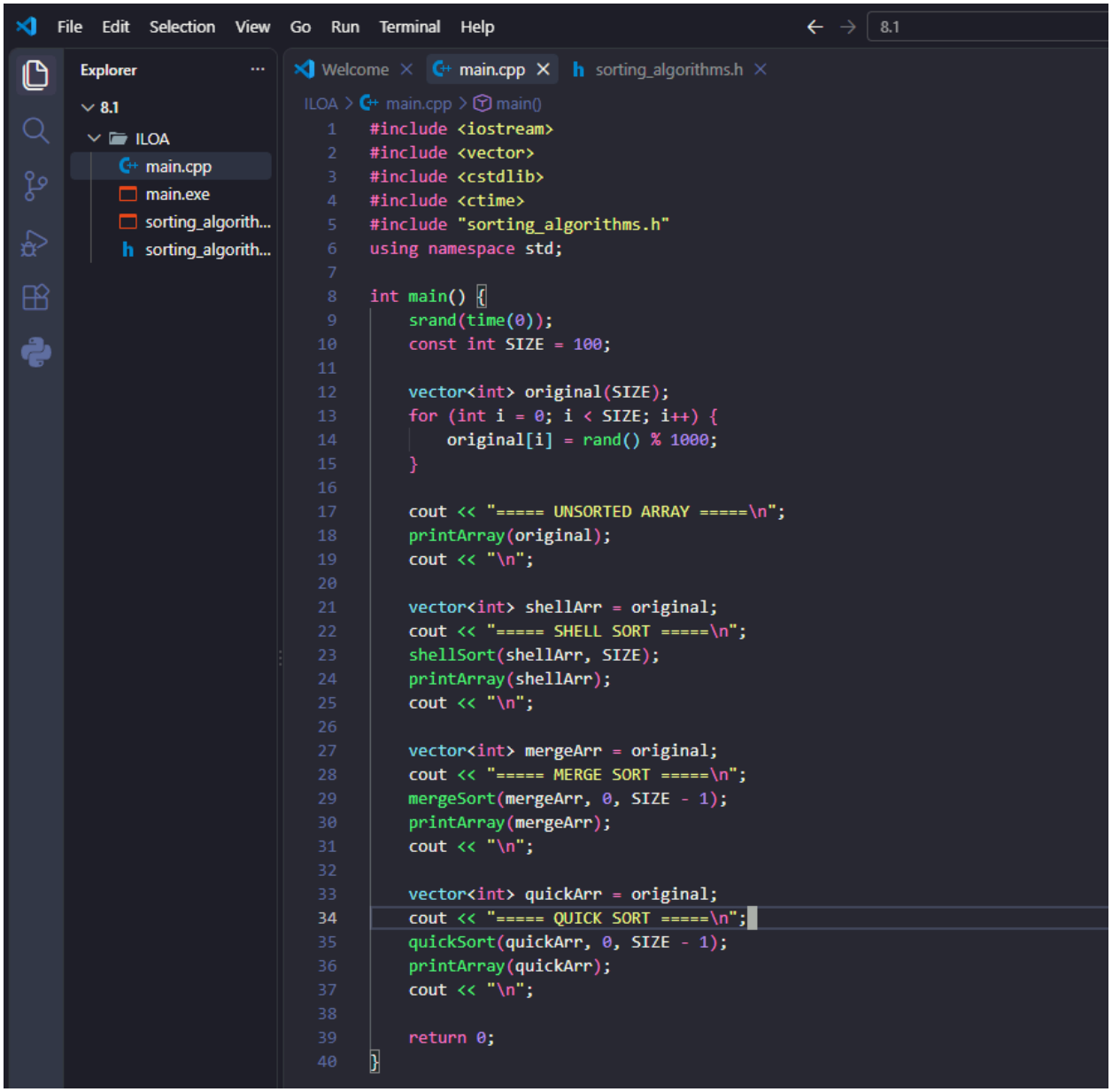
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ILO A

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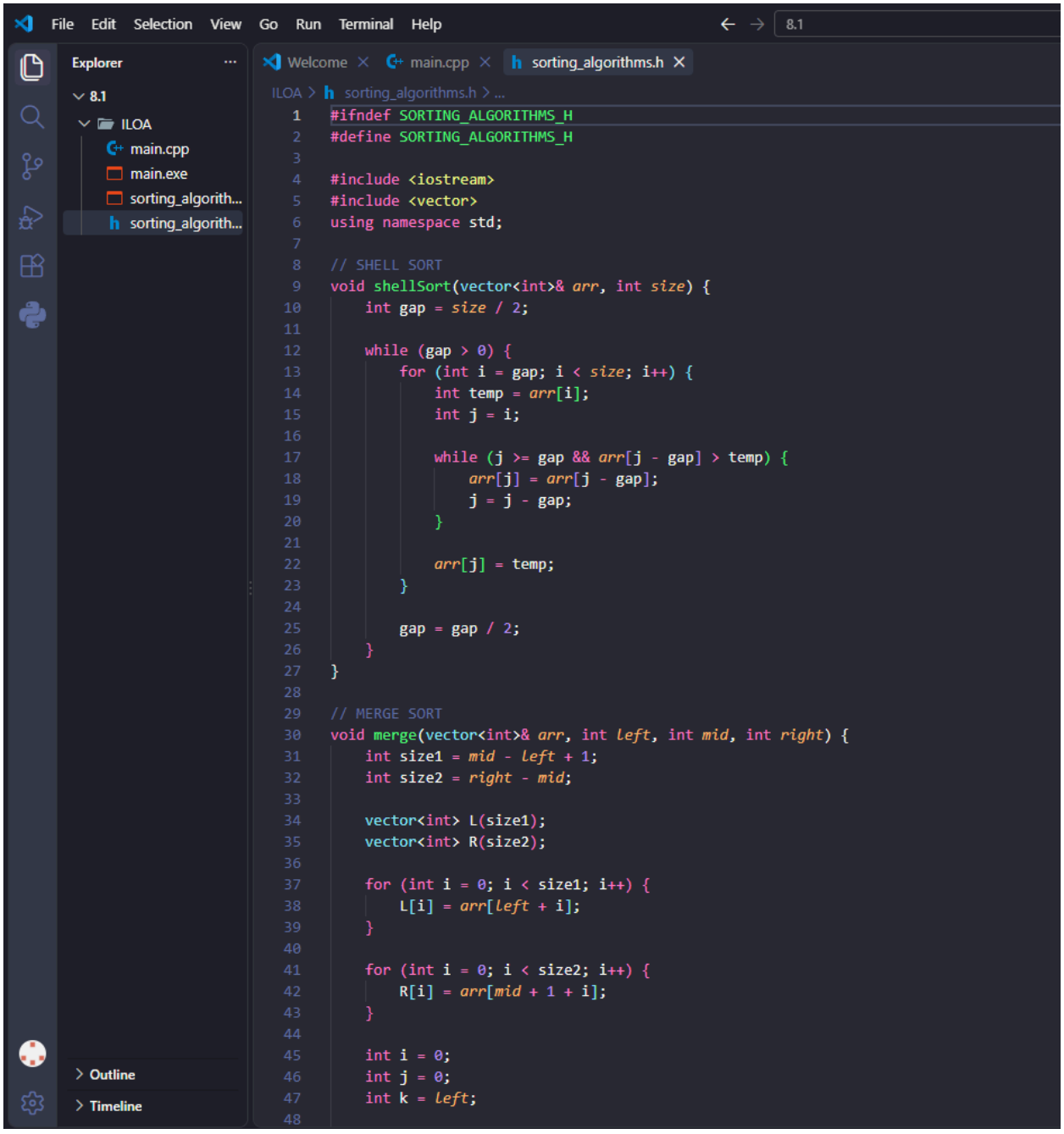
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```cpp
#ifndef SORTING_ALGORITHMS_H
#define SORTING_ALGORITHMS_H

#include <iostream>
#include <vector>
using namespace std;

// SHELL SORT
void shellSort(vector<int>& arr, int size) {
    int gap = size / 2;

    while (gap > 0) {
        for (int i = gap; i < size; i++) {
            int temp = arr[i];
            int j = i;

            while (j >= gap && arr[j - gap] > temp) {
                arr[j] = arr[j - gap];
                j = j - gap;
            }

            arr[j] = temp;
        }

        gap = gap / 2;
    }
}

// MERGE SORT
void merge(vector<int>& arr, int left, int mid, int right) {
    int size1 = mid - left + 1;
    int size2 = right - mid;

    vector<int> L(size1);
    vector<int> R(size2);

    for (int i = 0; i < size1; i++) {
        L[i] = arr[left + i];
    }

    for (int i = 0; i < size2; i++) {
        R[i] = arr[mid + 1 + i];
    }

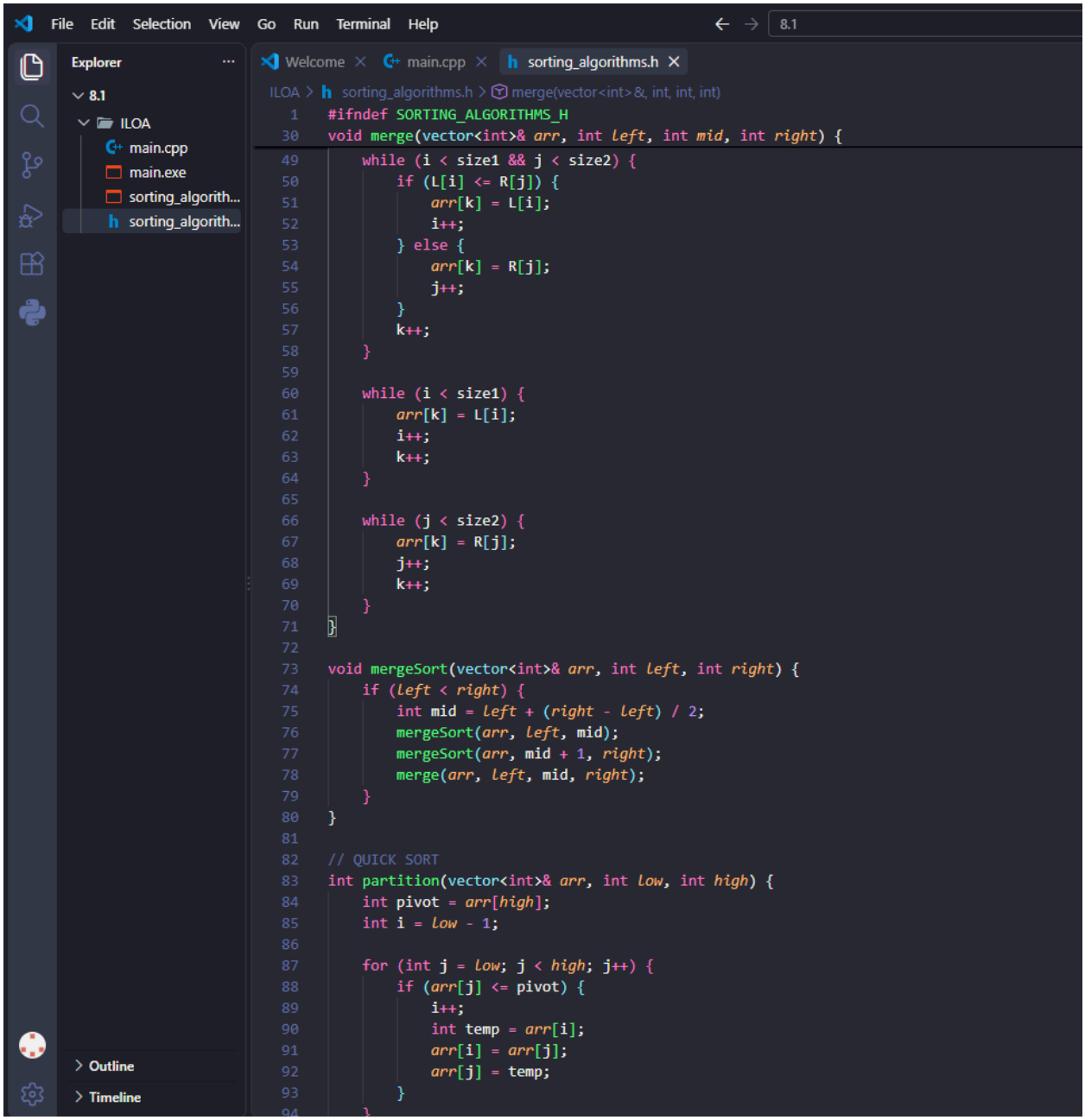
    int i = 0;
    int j = 0;
    int k = left;

```

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```
    int temp = arr[i + 1];
    arr[i + 1] = arr[high];
    arr[high] = temp;

    return i + 1;
}

void quickSort(vector<int>& arr, int low, int high) {
    if (low < high) {
        int p = partition(arr, low, high);
        quickSort(arr, low, p - 1);
        quickSort(arr, p + 1, high);
    }
}


void printArray(const vector<int>& arr) {
    for (int i = 0; i < arr.size(); i++) {
        cout << arr[i] << " ";
        if ((i + 1) % 10 == 0) {
            cout << endl;
        }
    }
    cout << endl;
}

#endif
```

OUTPUT

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File Edit Selection View Go Run Terminal Help

Explorer
8.1
ILOA
main.cpp
main.exe
sorting_algorit...
sorting_algorit...

Problems Output Debug Console Terminal Ports

```
[Done] exited with code=1 in 1.457 seconds

[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\Midt
===== UNSORTED ARRAY =====
183 629 99 125 106 948 179 798 692 220
619 168 209 197 144 964 109 955 753 428
966 808 539 162 318 420 918 398 729 330
296 783 298 572 913 541 802 305 303 516
443 958 555 642 145 275 491 796 171 408
129 662 537 631 4 347 206 763 917 44
392 738 869 44 517 71 981 755 889 603
590 272 349 805 739 21 50 871 155 902
887 643 939 330 918 614 140 526 85 956
272 590 900 676 747 253 691 5 695 318


===== SHELL SORT =====
4 5 21 44 44 50 71 85 99 106
109 125 129 140 144 145 155 162 168 171
179 183 197 206 209 220 253 272 272 275
296 298 303 305 318 318 330 330 347 349
392 398 408 420 428 443 491 516 517 526
537 539 541 555 572 590 590 603 614 619
629 631 642 643 662 676 691 692 695 729
738 739 747 753 755 763 783 796 798 802
805 808 869 871 887 889 900 902 913 917
918 918 939 948 955 956 958 964 966 981


===== MERGE SORT =====
4 5 21 44 44 50 71 85 99 106
109 125 129 140 144 145 155 162 168 171
179 183 197 206 209 220 253 272 272 275
296 298 303 305 318 318 330 330 347 349
392 398 408 420 428 443 491 516 517 526
537 539 541 555 572 590 590 603 614 619
629 631 642 643 662 676 691 692 695 729
738 739 747 753 755 763 783 796 798 802
805 808 869 871 887 889 900 902 913 917
918 918 939 948 955 956 958 964 966 981


===== QUICK SORT =====
4 5 21 44 44 50 71 85 99 106
109 125 129 140 144 145 155 162 168 171
179 183 197 206 209 220 253 272 272 275
296 298 303 305 318 318 330 330 347 349
392 398 408 420 428 443 491 516 517 526
537 539 541 555 572 590 590 603 614 619
629 631 642 643 662 676 691 692 695 729
738 739 747 753 755 763 783 796 798 802
805 808 869 871 887 889 900 902 913 917
918 918 939 948 955 956 958 964 966 981
```